

Mathematical Statistics

Lecture-4 The Birthday Problem in Probability

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Abstract

These notes cover the classic "Birthday Problem" in probability theory. We explore why our intuition often fails us in this scenario and introduce a powerful technique for solving complex probability problems: calculating the probability of the complementary event. The notes provide a step-by-step derivation of the solution, discuss computational challenges, and highlight the key takeaways and underlying assumptions.

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1 Introduction: An Unintuitive Problem

The lecture begins by introducing a classic, and surprisingly unintuitive, problem in probability theory: **The Birthday Problem**.

1.1 The Problem Statement

The Question

If there are n people in a room, how large does n have to be for there to be at least a 50% chance that two or more people share the same birthday?

1.2 Common Intuition vs. Reality

A common off-the-cuff guess is that you would need about half the number of days in a year, something like $365/2 \approx 183$ people. This intuition is based on filling up "slots" (days of the year), but it fails to account for the rapidly growing number of *pairs* of people being compared. As we will see, the actual answer is much smaller and demonstrates a key principle in probability.

2 The Core Strategy: The Power of the Complement

Directly calculating the probability of "at least two people sharing a birthday" is extremely complex. This event could mean exactly one pair shares a birthday, or exactly three people, or two different pairs share two different birthdays, and so on. This would require calculating the probability of each specific scenario and adding them up—a messy and error-prone process.

💡 Key Concept

A much simpler approach is to calculate the probability of the **complementary event**. The complement of "at least two people share a birthday" is "**no one shares a birthday**".

The Law of Total Probability states that the probability of an event happening plus the probability of it not happening must equal 1 (or 100%). Therefore, we can use the following formula:

$$P(\text{at least 2 share}) = 1 - P(\text{no one shares})$$

This transforms our problem into finding the probability that every single person in the room has a unique birthday, which is a much easier task to calculate.

3 Calculating P(no one shares a birthday)

Let's calculate the probability that in a room of n people, all birthdays are unique. We'll make two key assumptions for this calculation:

1. There are 365 days in a year (we ignore leap years).
2. Each day of the year is an equally likely birthday for any person (a uniform distribution).

We can think about this problem sequentially, person by person.

- **Person 1:** Can have a birthday on any of the 365 days. Since there are no other people to clash with, the probability of them having a unique birthday is certain.

$$P(\text{Person 1 unique}) = \frac{365}{365}$$

- **Person 2:** For Person 2 to have a unique birthday, their birthday must fall on one of the 364 days not taken by Person 1.

$$P(\text{Person 2 unique} \mid \text{Person 1 unique}) = \frac{364}{365}$$

- **Person 3:** For Person 3 to have a unique birthday, their birthday must fall on one of the 363 days not taken by the first two people.

$$P(\text{Person 3 unique} \mid \text{First 2 unique}) = \frac{363}{365}$$

- **Person n:** Following this pattern, for the n -th person to have a unique birthday, their birthday must not match any of the previous $n - 1$ unique birthdays.

$$P(\text{Person } n \text{ unique} \mid \text{First } n-1 \text{ unique}) = \frac{365 - (n - 1)}{365} = \frac{365 - n + 1}{365}$$

Since these are a series of independent events, the total probability that *no one* shares a birthday is the product of all these individual probabilities:

$$P(\text{no one shares}) = \frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \cdots \times \frac{365 - n + 1}{365}$$

4 An Alternative View: Using Permutations and Combinatorics

We can also frame this problem using combinatorics, which gives us a more compact formula.

1. **Total Possible Outcomes:** Each of the n people can have their birthday on any of the 365 days. The total number of ways to assign birthdays to n people is:

$$\text{Total Arrangements} = 365 \times 365 \times \cdots \times 365 = 365^n$$

2. **Favorable Outcomes (No Shared Birthdays):** This is the number of ways to assign a unique birthday to each of the n people. This is a permutation problem: choosing n distinct days from 365 and assigning them to n people.

$$\text{Number of Unique Arrangements} = P(365, n) = \frac{365!}{(365 - n)!}$$

3. **The Probability:** The probability of no one sharing a birthday is the ratio of favorable outcomes to total outcomes.

$$P(\text{no one shares}) = \frac{\text{Number of Unique Arrangements}}{\text{Total Arrangements}} = \frac{\frac{365!}{(365 - n)!}}{365^n}$$

Both the sequential product and the permutation formula will yield the exact same result.

5 Solving the Problem and Computational Nuances

Our goal is to find the smallest n for which:

$$P(\text{at least 2 share}) > 0.5$$

Which is equivalent to:

$$1 - P(\text{no one shares}) > 0.5 \implies P(\text{no one shares}) < 0.5$$

Important Note

The factorial formula ($365!$) involves incredibly large numbers. '365!' is so large that most standard calculators and programming languages will return an overflow error. To compute this, it's best to use the iterative product formula in a loop. Start with a probability of 1 and multiply by $\frac{364}{365}$, then $\frac{363}{365}$, and so on, checking after each step when the result drops below 0.5.

By performing this calculation, we find that:

- For $n = 22$, $P(\text{at least 2 share}) \approx 47.6\%$
- For $n = 23$, $P(\text{at least 2 share}) \approx 50.7\%$

Therefore, you only need **23 people** in a room to have a greater than 50% chance that at least two of them share a birthday.

6 Summary and Key Takeaways

- **Unintuitive Results:** The Birthday Problem is a powerful illustration that our intuition can be misleading in probability. The probability grows much faster than we expect because the number of possible pairs of people, given by $\frac{n(n-1)}{2}$, grows quadratically.
- **The Complement Rule:** When a probability calculation seems overwhelmingly complex, always consider if calculating the probability of its complement ($1 - P(\text{not A})$) is simpler. It often is.
- **Assumptions Matter:** Our solution is based on simplifying assumptions (365 days, uniform distribution). In reality, birthdays are not uniformly distributed. This slight non-uniformity would actually make shared birthdays *more* likely, meaning the number of people required might be even less than 23.

Exercise Problem

The "Username Collision" Problem

A new social media platform generates a random 4-character username for each new user. Each character can be any lowercase letter (a-z) or a digit (0-9). This gives a total of 36 possible options for each character.

1. How many unique 4-character usernames are possible?
2. Using the principles from the Birthday Problem, calculate the minimum number of users (n) that need to sign up before the probability of at least two users being assigned the same username (a "collision") exceeds 50%.

Hint: Think of the total number of unique usernames as the "number of days in the year" from the original problem.

Problem Statement

A new social media platform generates a random 4-character username for each new user. Each character can be any lowercase letter (a-z) or digit (0-9), so there are 36 possible choices per character.

1. How many unique 4-character usernames are possible?
2. Using the analogy to the Birthday Problem, calculate the minimum number of users (n) who need to sign up before the probability of at least two users being assigned the same username (a "collision") exceeds 50%.

Solution

1. Total Unique Usernames

Each character can be selected in 36 ways. Number of possible usernames:

$$N = 36^4 = 1,679,616$$

So, there are **1,679,616** unique usernames possible.

2. Username Collision Probability (Birthday Problem)

Let $N = 36^4$ be the total number of unique usernames. Let n be the number of users. The probability that all n usernames are unique:

$$P(\text{no collision}) = \prod_{k=0}^{n-1} \left(1 - \frac{k}{N}\right)$$

For large N , this can be approximated using the exponential function:

$$P(\text{no collision}) \approx \exp\left(-\frac{n(n-1)}{2N}\right)$$

The probability of at least one collision is:

$$P(\text{collision}) = 1 - P(\text{no collision})$$

We want the smallest n such that $P(\text{collision}) > 0.5$:

$$\begin{aligned}P(\text{no collision}) &< 0.5 \\ \exp\left(-\frac{n^2}{2N}\right) &< 0.5 \\ -\frac{n^2}{2N} &< \ln(0.5) = -\ln(2) \\ \frac{n^2}{2N} &> \ln(2) \\ n^2 &> 2N \ln(2) \\ n &> \sqrt{2N \ln(2)}\end{aligned}$$

Plug in the values:

$$\begin{aligned}N &= 36^4 = 1,679,616 \\ \ln(2) &\approx 0.6931 \\ n &> \sqrt{2 \times 1,679,616 \times 0.6931} \approx \sqrt{2,327,557} \approx 1,525\end{aligned}$$

Thus, the minimum number of users required for the collision probability to exceed 50% is:

$$n = 1,526$$

Final Answers

- **Total unique usernames available:** $36^4 = 1,679,616$
- **Minimum users for $> 50\%$ collision probability:** $n = 1,526$

7 Introduction: The Quality Control Problem

In many manufacturing settings (e.g., producing aircraft parts, electronics, or pharmaceuticals), it's crucial to ensure the quality of the products. A key challenge is that testing a product's quality often involves **destructive inspection**—the item being tested must be broken, cut open, or tested to failure to verify its integrity.

Real-World Applications

This presents a dilemma: How can a factory assess the quality of a large batch (or "lot") of items without destroying all of them in the process?

The solution lies in **statistical sampling**. By taking a small, random sample from the larger lot and testing it, we can make an informed inference about the quality of the entire lot. This lecture explores the probability behind this process.

8 Framing the Probabilistic Question

Let's define the variables for our quality control scenario:

- **N**: The total number of items in the lot (e.g., in a warehouse).
- **K**: The total number of *defective* items within the entire lot N.
- **R**: The number of items we randomly select for our sample.
- **M**: The number of *defective* items we find in our sample R.

The Core Question

Given a lot of size N containing K defective items, what is the probability of finding exactly M defective items in a random sample of size R?

To answer this, we will use a fundamental principle of probability:

$$P(\text{Event}) = \frac{\text{Number of ways the event can happen}}{\text{Total number of possible outcomes}}$$

9 Breaking Down the Populations

To count the outcomes correctly, we first need to categorize the items in both the total lot and our sample.

• The Total Lot (N items):

- Defective items = **K**
- Good items = **N - K**

• Our Sample (R items):

- Defective items = **M**
- Good items = **R - M**

The M defective items in our sample must come from the K defective items in the lot, and the R - M good items in our sample must come from the N - K good items in the lot.

10 The Mathematical Tool: Combinations

When we take a sample from a lot, we are performing **sampling without replacement**. Furthermore, the **order** in which we select the items for our sample does not matter.

💡 Key Concept

Definition: Combination

Whenever we are counting the number of ways to choose a subset of items where order is irrelevant, we use the **combination** formula, often read as "n choose k". The number of ways to choose a subset of *k* items from a total set of *n* items is given by:

$$\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$$

where *n*! (n factorial) is the product of all positive integers up to n (e.g., $4! = 4 \times 3 \times 2 \times 1 = 24$).

11 Deriving the Probability Formula Step-by-Step

Now, let's apply the combination formula to our probability framework.

1. Calculate the Total Number of Possible Outcomes (The Denominator)

First, we need to find the total number of unique samples of size R that we could possibly draw from the lot of N items.

$$\text{Total possible samples} = \binom{N}{R}$$

2. Calculate the Number of Favorable Outcomes (The Numerator)

Next, we calculate the number of ways our specific event—getting exactly M defective items and $R - M$ good items—can occur. This is a two-part process:

- *Part A: Choosing the Defective Items:* The number of ways to choose M defective items from the K total defective items is:

$$\text{Ways to choose defective items} = \binom{K}{M}$$

- *Part B: Choosing the Good Items:* The number of ways to choose the remaining $R - M$ items from the $N - K$ total good items is:

$$\text{Ways to choose good items} = \binom{N - K}{R - M}$$

Since these choices are independent, we multiply them to get the total number of favorable outcomes:

$$\text{Total favorable outcomes} = \binom{K}{M} \times \binom{N - K}{R - M}$$

3. Assemble the Final Formula

By dividing the favorable outcomes by the total outcomes, we get the probability:

$$P(M \text{ defective in sample}) = \frac{\binom{K}{M} \binom{N - K}{R - M}}{\binom{N}{R}}$$

Problem Statement

A factory produces a total of $n = 1000$ parts in one batch. Out of these, $K = 50$ parts are defective. A non-destructive sampling inspection is performed by selecting $R = 20$ parts (without replacement) at random. We want to find the probability that exactly $M = 2$ of the selected parts are defective.

Step 1: Statistical Model

Since the sampling is conducted without replacement from a finite population, the number of defective parts X follows the **hypergeometric distribution**, which is a special case of the multinomial distribution.

$$P(X = M) = \frac{\binom{K}{M} \binom{n-K}{R-M}}{\binom{n}{R}}$$

where

- n : total number of items in the lot,
- K : number of defectives in the lot,
- R : number of items drawn in the sample,
- M : number of defectives found in the sample.

Step 2: Substitute Known Values

$$P(X = 2) = \frac{\binom{50}{2} \binom{950}{18}}{\binom{1000}{20}}$$

Step 3: Simplify the Combinatorial Terms

$$\binom{50}{2} = \frac{50 \times 49}{2} = 1225$$

Hence,

$$P(X = 2) = \frac{1225 \times \binom{950}{18}}{\binom{1000}{20}}$$

Step 4: Approximate Numerical Value

Computationally (using statistical software or a calculator):

$$P(X = 2) \approx 0.27$$

Thus, there is approximately a **27% probability** that exactly two defective parts appear in the 20-part sample.

Problem 2: Quality Control Inspection

A manufacturing company produces a batch of $n = 1000$ parts, among which $K = 50$ are defective. To evaluate product quality, an engineer selects a sample of $R = 20$ parts at random, without replacement. Let M denote the number of defective parts found in the sample.

Question: What is the probability that exactly $M = 2$ defective parts are found in the sample?

Problem 2: General Population Sampling

In a large population of N items, there are B defective and $G = N - B$ non-defective items. A random sample of n items is drawn without replacement. In the sample, b defective and $g = n - b$ non-defective items are observed.

Question: Derive the general expression for the probability

$$P(X = b)$$

that exactly b defective items appear in the sample, and show how this expression relates to the multinomial form for two categories (good vs. defective).

Step 5: Interpretation

This model allows engineers to determine the likelihood of finding a given number of defective parts in a small, non-destructive sample—essential for quality assurance. When the number of defectives in the full population (K) is unknown, we can use the observed sample result (M) to estimate K through inference methods such as confidence intervals or Bayesian estimation.

Step 6: Generalization (Multivariate Case)

For quality inspections with multiple defect types (e.g., minor, major, and critical), the **multivariate hypergeometric distribution** applies:

$$P(x_1, x_2, \dots, x_k) = \frac{\prod_{i=1}^k \binom{K_i}{x_i}}{\binom{n}{R}}$$

subject to $\sum_{i=1}^k x_i = R$ and $\sum_{i=1}^k K_i = n$.

Step 7: Final Expression

$$P(X = 2) = \frac{\binom{50}{2} \binom{950}{18}}{\binom{1000}{20}} \approx 0.27$$

12 The Hypergeometric Distribution

This formula defines an important probability distribution known as the **Hypergeometric Distribution**.

💡 Key Concept

The Hypergeometric Distribution

Describes the probability of k successes (e.g., defective items) in n draws, **without replacement**, from a finite population of size N that contains exactly K objects with that feature.

Note on Terminology: While related, this is more precisely called the Hypergeometric distribution. The Multinomial distribution is a generalization of the Binomial distribution for more than two categories, typically assuming sampling *with* replacement. The key feature here is sampling **without replacement**.

13 Looking Ahead: The Inverse Problem of Statistics

The formula we derived is a tool of **probability**. It assumes we know the properties of the overall population (i.e., we know N and K).

In a real-world quality control scenario, **we often do not know K** . The entire point of sampling is to *estimate* the unknown number of defective items in the lot. This leads to the **inverse problem**, which is the core of **statistics**:

Given the results from our sample (we know R and M), what is the most likely value, or range of values, for the unknown K ?

Answering this question involves statistical inference, and techniques like Bayesian statistics (using Bayes' Theorem) are designed to solve such inverse problems by updating our beliefs about an unknown quantity based on observed data.

14 Summary and Key Takeaways

- **Real-World Application:** Probability is essential for practical applications like industrial quality control, allowing for efficient inspection without testing every item.
- **The Right Tool:** The Hypergeometric distribution is the correct model for calculating probabilities when sampling **without replacement** from a finite population.
- **Probability vs. Statistics:**
 - **Probability** calculates the chance of future events based on known parameters. (e.g., "If I know K , what's the chance of finding M ?")
 - **Statistics** uses data from observed events to infer the unknown parameters of a system. (e.g., "I found M , so what is the most likely value of K ?")

Exercise Problem

A Deck of Cards

A standard deck contains 52 cards, of which 12 are "face cards" (Jack, Queen, King). Suppose you are dealt a 5-card poker hand.

1. What is the total number of possible 5-card hands you can be dealt? (This is your denominator).
2. What is the probability that your hand contains **exactly 2 face cards**?
3. What is the probability that your hand contains **no face cards**?